

CSA0619-Design and Analysis of Algorithm

28. Debugging Pair Sum Code (Duplicate and Mirror Pair Detection):

main.py	Run	Output
<pre>1 # Debugged and Optimized Pair Sum Code 2 # Removes duplicate and mirror pairs 3 def pair_sum_all(arr, target): 4 pairs = [] 5 # Check each pair only once 6 for i in range(len(arr)): 7 for j in range(i + 1, len(arr)): 8 # Check if pair sum equals target 9 if arr[i] + arr[j] == target: 10 pairs.append((i, j)) 11 return pairs 12 # Main Program 13 arr = list(map(int, input("Enter array elements: ").split())) 14 target = int(input("Enter target sum: ")) 15 result = pair_sum_all(arr, target) 16 print("Valid Pairs:", result)</pre>		<pre>Enter array elements: 2 4 6 8 Enter target sum: 10 Valid Pairs: [(0, 3), (1, 2)] === Code Execution Successful</pre>

Analysis:

The duplicate pairs occur because the inner loop starts from 0. Therefore, every valid pair is checked twice in reverse order, leading to redundant comparisons and repeated output.

Optimization:

To remove duplicate and mirror pairs, the inner loop should start from $i + 1$ instead of 0. This ensures that each pair is checked only once without missing any valid solution.

Report Analysis

Case	Time Complexity
Best	$O(n^2)$
Average	$O(n^2)$
Worst	$O(n^2)$

Space Complexity: $O(k)$

Where:

- **n = Number of elements in the array**
- **k = Number of valid pairs stored**

